SRIKALA SANDU

New Jersey (open to relocation) | (551) 554-1160 | sandusrikala@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

AI/ML engineer and researcher specializing in generative AI, backend systems, and computer vision. Cut RAG retrieval latency ~30%, owned the AI and backend for a 10,000+ user social app, won HR Hackathon 2026, and co-authored an ACM ASSETS 2025 paper. Skilled in Python, LangChain, Hugging Face, OpenCV, Node.js, and AWS.

EDUCATION

Stevens Institute of Technology — **Hoboken, NJ**
Master of Science in Computer Science, with Honors (GPA: 3.8/4.0) | Provost Scholarship — May 2026
- Coursework: Advanced Algorithm Design, Mathematical Foundations of ML, HCI, Data Analytics & ML, Cybersecurity

Visvesvaraya Technological University — **Bangalore, India**
Bachelor of Engineering in Computer Science (GPA: 8.0/10) — Aug 2021
- Teaching assistant for Advanced Algorithm Design and Data Structures; final project: Stock Trend Forecasting (85% accuracy)

SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, C++, C, Java, Kotlin, R, PHP
Machine Learning & AI: PyTorch, scikit-learn, XGBoost, OpenCV, NetworkX, LangChain, LangGraph, Hugging Face, RAG, RAGAS, DeepEval, LLM evaluation, prompt engineering, A/B testing
Web & Mobile: React.js, React Native, Expo, Node.js, Express, FastAPI, REST APIs, Android (Camera2 API)
Tools & Technologies: Git, Docker, AWS, Linux/Unix, Twilio, ElevenLabs, PRAW, Jupyter
Database: MongoDB, MySQL, Firebase, Firestore, Supabase, Pinecone

EXPERIENCE

Sweeesh | Co-founder — **Sep 2026 – Present**
- Co-founded Sweeesh after our team won HR Hackathon 2026 (NYU Tandon) with a prototype that tackles unverifiable resume claims
- Implemented Ed25519 signatures and delegation-chain verification so employers can trust endorsements (Node.js, Express)

G5 Infotech | Generative AI Engineer — **Mar 2026 – Apr 2026**
- Designed a pipeline benchmarking 3+ RAG chunking strategies (fixed-size, semantic, recursive) with LangChain and Hugging Face
- Measured each strategy on precision, recall, and MRR; adopting recursive chunking cut retrieval latency ~30% versus the previous setup
- Tuned Pinecone top-k and context-window settings in Python to curb token overhead, reducing average response time ~22%

Stevens Institute of Technology | Graduate Research Assistant — **Jan 2025 – Jun 2026**
- Developed an Android app for Vuzix M400 smart glasses that lets blind users fill printed forms independently by voice (AWS)
- Programmed real-time field detection (Kotlin, OpenCV 4.9, Camera2) with Canny edges, contours, and perspective warp
- Evaluated forms from 25 blind participants (manual review and IoU): 87% of fields mappable and 68% more efficient than baseline
- Produced Likert usability analyses in R and co-authored the resulting ACM ASSETS 2025 paper

CircleZapp | Software Developer — **Dec 2024 – Feb 2025**
- Owned the data science, AI features, and backend (Python, NetworkX, Firebase) for a social app with 10,000+ active accounts
- Engineered matching bots and a questionnaire that score user compatibility by interests, location, school, and community
- Cut match latency from 800ms to 340ms under high input volume and A/B-tested 3 recommendation variants, lifting click-through 12%

Reddit Inc. | AI Security Engineer — **Apr 2023 – Aug 2024**
- Deployed Python detection bots (Reddit API, PRAW) that flagged harmful and abusive content in real time at 92% accuracy
- Lowered toxic-behavior reports 35% and cut manual moderation workload 40% for a team of 8 moderators
- Authored runbooks that shortened moderator onboarding from ~3 weeks to ~10 days

Certes Networks | Backend Developer — **Apr 2022 – Apr 2023**
- Shipped C++ features for encryption hardware (Syslog, LM Sensors) and a low-level LCD Linux driver in C, using Docker
- Refactored 5+ high-throughput modules, improving encryption efficiency 20% while sustaining 99.8% uptime
- Diagnosed silent encryption failures while testing legacy CEPs on a new device and resolved them by upgrading to AES-256

PROJECTS

MindMate – AI Emotional Companion | GitHub | QuacksHacks (MLH), Stevens Institute of Technology — **2026**
- Created an AI wellness companion with chat, voice, and Twilio phone check-ins, using GPT-4 structured outputs for daily micro-actions (React, Node.js, Firebase, ElevenLabs)

Sprout – Adaptive Learning App | GitHub | TRAE.ai × MiniMax NYC Hackathon — **2026**
- Launched a React Native/Expo children's learning app with an AI adaptive-difficulty engine and Supabase (team of 4)

AI College Admission Chatbot | GitHub
- Automated admission inquiries with a Python NLP chatbot, answering at 95% accuracy and improving efficiency 40%

PUBLICATION

R. R. Khan, **S. Sandu**, J. Hong. Interactive Form Filling Assistant on Smart Glasses for Blind Users. ACM ASSETS. — **Oct 2025**